Common-Region Skeleton–Surface Encoding for Scale-Aware Neuronal Morphology Analysis

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Abstract

Neuronal morphology analysis supports reconstruction proofreading, connection completion, and cell characterization from local neurites to whole cells. Jointly learning from skeletons and surfaces faces two linked obstacles: limited sampling can omit geometry and obscure size differences, while object-level fusion lacks an explicit association between a branch and its surrounding surface. We formulate these obstacles through measure-based geometric representation: sampling determines which geometry is represented and with what weight, and regional aggregation determines where heterogeneous evidence is combined. Guided by this formulation, we propose the Common-Region Skeleton–Surface (CRSS) framework with two mechanisms. Weighted multiresolution geometric encoding combines global and local surface samples with the observed skeleton and position/length embeddings. Topology-guided common-region fusion separately aggregates skeletal and surface evidence on shared arclength regions and learns joint tokens before contextual propagation and task readout. We evaluate the framework on five tasks from NREDD, H01, and FlyWire using matched development-set protocols with seed 42. At the same point budget, the sampling mechanism improves FlyWire macro-F1 by 9.91 percentage points over uniform sampling; learned regional interaction improves H01 local and FlyWire classification by 1.92 and 8.90 points over regional averaging. These results support connecting input coverage with explicit local correspondence for neuronal morphology learning.

1. Introduction

Large-scale connectomics now provides detailed neuronal reconstructions from human cortex and whole fly brains, alongside functionally characterized mouse circuits [10, 26, 29]. Turning these resources into biological evidence requires identifying reconstruction errors, completing connections, characterizing compartments, and predicting cell attributes. These decisions span local neurites and entire arbors. Skeletons expose observed connectivity, while surfaces retain shape around the centerline; a useful morphology encoder must preserve and relate both kinds of evidence across these spatial extents.

Recent neuronal graph models learn local and global structure, positional priors, and multimodal representations [25, 27, 28, 43]. NEURD already relates meshes to branch graphs, and segment-connection prediction combines global morphology with local EM appearance [3, 6]. Our question is therefore specific: under a finite observation budget, how can an observed skeleton and its co-registered three-dimensional surface form shared local representations for multiple prediction granularities?

Two linked difficulties arise. *Evidence retention* concerns what reaches the encoder: a fixed global sample can poorly cover small branches in a large arbor, while object-wise unit scaling can erase size differences. Sampling allocation, density, and physical extent are distinct decisions [33, 42]. *Evidence association* concerns how retained observations interact: skeletal nodes and surface patches have different discretizations and no one-to-one correspondence. Whole-object fusion does not explicitly assign a surface variation to a branch, and nearby branches complicate purely spatial association. Missing input evidence cannot be restored by fusion; sufficient coverage alone does not establish correspondence.

We organize both difficulties through a *weighted geometric representation*: each element records a location and the amount of source geometry it represents, and regional features are normalized local aggregates. This formulation separates sampling from aggregation support. It motivates retaining distributed observations and size cues, then combining skeleton and surface evidence on common regions while normalizing their different geometric quantities separately. This is a mathematical basis for the design, rather than a claim of sampling or scale invariance.

Accordingly, we propose the *Common-Region Skeleton–Surface* (CRSS) framework. Its *weighted multiresolution geometric encoding* combines global and local surface sam-

ples, retains the observed skeleton, and embeds regional position and length. Its *topology-guided common-region fusion* associates surface patches with skeletal arclength regions, aggregates each modality separately, and learns joint regional tokens. These tokens enter regional context propagation before local, pairwise, or whole-cell readout. The two mechanisms thus connect evidence retention to local interaction; the architectural change concerns both the unit and the placement of fusion.

Five separately supervised tasks on NREDD, H01, and FlyWire evaluate this design under matched development protocols with seed 42. Equal-budget sampling improves FlyWire macro-F1 by 9.91 percentage points over uniform sampling; learned interaction improves H01 local and FlyWire macro-F1 by 1.92 and 8.90 points over regional averaging. The contributions are:

- **Weighted multiresolution inputs** that retain spatial coverage and size cues through sampling weights and position/length embeddings.
- **Common-region joint encoding** that forms skeleton–surface tokens before contextual propagation and task-specific readout.
- **Matched mechanism evaluations** across five tasks, including equal-budget sampling and correspondence perturbations, with task-dependent effects.

2. Related Work

Connectomic resources and analysis. Human, larval-fly, adult-fly, and functional mouse connectomes expose morphology at increasing scale [10, 26, 29, 32]. Hierarchical fly annotations, a visual-system inventory, and compartment-specific inhibitory organization connect morphology to cell identity and circuit roles [19, 23, 24]. Gapr and CAVE support collaborative reconstruction and reproducible annotation, whereas NEURD decomposes meshes into annotated branch graphs [3, 11, 14]. These works motivate local and cell-level analysis; their full resource sizes are distinct from our task subsets.

Learned neuronal representations. MACGNN, MorphoGNN, MorphRep, and DSM learn representations from graphs or projection trajectories [12, 37, 43, 44]. SGT-Morph combines local topology and global reasoning; NeuroGT adds biophysical and hierarchical encodings; NeuNet integrates single-cell skeletons with circuit connectivity [17, 27, 28]. GraPHFormer explores late fusion and cross-attention between skeletons and persistence images [25]. Multimodal segment-connection prediction uses morphology and local EM images, while MoGen generates neuronal fragment point clouds for shape modeling and proofreading augmentation [6, 21]. Our setting instead learns from an observed skeleton and an independently sampled, co-registered surface, with explicit regional association rather than assuming that multimodality itself is new.

Point geometry and observation budgets. PointVector, PointMeta, and KPConvX refine local feature aggregation [9, 18, 30]; PTv3, Flash3D, and Superpoint Transformer organize efficient spatial context [5, 22, 34]. APES selects salient samples, while GeoMAE and Sonata study the geometric information encouraged by pretraining [31, 33, 35]. LitePT separates local geometric processing from broader attention, and MSPT combines local and coarse interactions [8, 40]. Utonia distinguishes scale, density, and sampling differences across domains [42]. These precedents motivate separating input coverage from receptive field: broader attention cannot access observations omitted during sampling. Our contribution concerns allocating and weighting observations for joint skeletal–surface regions, not a new point-attention operator.

Regional correspondence and geometric priors. Pi-MAE couples image and point representations through cross-modal reconstruction [4]. For registration, 2D3D-MATR and Hg-I2P organize heterogeneous correspondences, while RoITr, PEAL, and SuP address pose, overlap, or sub-cloud matching [2, 13, 16, 38, 39]. FusionCell lets topology query layout geometry [41]. These mechanisms seek or use correspondence under different input assumptions; our inputs already share a coordinate frame, and regions organize their unequal geometric supports for prediction.

CAJAL compares cellular morphologies through metric geometry, whereas VesselTok learns centerline-conditioned tokens for tubular anatomy [15, 20]. LEAP encodes local graph structure, while URoPE uses camera geometry for cross-view positional relations [1, 36]. Our supports are already co-registered: we preserve their observed connectivity and size information while learning regional content. These different settings distinguish geometric priors from a guarantee of resampling or physical-scale invariance.

3. Preliminaries

Inputs and prediction granularity. An example contains an observed skeleton $G = (V, E, \mathbf{x})$, surface samples $P = \{(\mathbf{p}_i, w_i)\}_{i=1}^{n}$ with $w_i > 0$, and an optional query q. A query denotes a location or a node pair; a whole-cell task uses the complete support. Task labels supervise prediction but never construct the input regions. In particular, target missing edges are excluded from E. Skeleton and surface coordinates share a frame within each example. We retain dataset-specific units: H01/FlyWire use micrometers, while NREDD retains its native-coordinate calibration limitation.

Weighted regional representation. For modality $m \in \{G, P\}$, region u receives nonnegative incidence weights A^m_{ui} over geometric elements x_i. Write the finite measure and its normalized feature aggregate as

$$\mu^m_u = \sum_i A^m_{ui}\delta_{x_i},$$
$$\mathcal{A}^m_u(f^m_u) = \frac{\int f^m_u \, d\mu^m_u}{\max(\epsilon, \int 1 \, d\mu^m_u)} = \frac{\sum_i A^m_{ui} f^m_{ui}}{\max(\epsilon, \sum_i A^m_{ui})}. \quad (1)$$

Here δ_{x_i} is a unit point mass, f^m_{ui} may include region-relative features, and $\epsilon > 0$ prevents division by zero. Sampling determines available elements and source weights; association determines their regional allocation. Each modality is normalized separately because skeletal length and surface sampling mass are different quantities. Region position and length retain support geometry. Equation 1 defines the implemented weighted sums; no convergence or sampling-invariance theorem is assumed.

4. Common-Region Skeleton–Surface Encoder

4.1. Weighted multiresolution geometric encoding

For whole-cell inputs, a global thumbnail is combined with spatially partitioned local surface samples, following the tile/thumbnail organization used in InternVL [7]. The H01 empirical-point sampler chooses up to 4096 global points and 2048 points per tile, with $K = \min(32, \max(1, \lceil N/16384 \rceil))$ tiles for N source points. Tiles are formed by splitting along the axis of largest spatial extent. If the global sample contains n_0 points and tile k contains N_k source points with n_k sampled, their weights are

$$w_i = \begin{cases} N/(2n_0), & i \text{ in the thumbnail}, \\ N_k/(2n_k), & i \text{ in tile } k. \end{cases} \quad (2)$$

The two views jointly retain total empirical mass N; repeated coordinates have their weights combined. This is a sampling-weight construction, not a claim that nonlinear network outputs are invariant to resampling. FlyWire uses its frozen surface-area sampling weights; local tasks retain their local support. The observed skeletal nodes and edges are preserved within each task's input support.

The input embeddings use node-to-region displacement, patch-to-polyline distance and axial position, centered region coordinates, and region length. The retained scale MLP receives $(\log(1+\ell_u), \log(1+r), \log(1+\delta_0))$, where r and $\delta_0 = 0.5$ are fixed in the retained configuration. Thus the variable scale input is primarily region length, not a learned per-tile resolution. Lengths and coordinates follow each dataset's declared unit convention; H01 and FlyWire use micrometers, while NREDD retains its native-coordinate calibration limitation. We preserve size cues rather than claim arbitrary-scale invariance. Utonia provides a related distinction between perceptual granularity and coordinates [42].

4.2. Topology-guided common-region fusion

We split non-branching chains of the observed skeleton into approximately fixed-arclength regions, retaining the actual polyline pieces rather than replacing a curved region by its chord. Let c_u and ℓ_u be the center and length of region u, and let r be the fixed support reference. A surface patch j has center c_j and mass M_j. Its soft association is

$$a_{uj} = \mathbf{1}[d(c_j, R_u) \le 2r] \exp\left[-\frac{d(c_j, R_u)^2}{2r^2}\right],$$
$$\pi_{uj} = \frac{a_{uj}}{\sum_v a_{vj}}, \qquad A^P_{uj} = \pi_{uj} M_j. \quad (3)$$

The denominator is taken only for supported patches. A patch without skeletal support becomes a surface-only region. Skeletal weights A^G_{ui} allocate the arclength of each observed segment piece to its endpoint features by linear interpolation. This defines common indices without assuming one-to-one node–patch correspondence. Related precedents include topology-guided geometry queries in FusionCell and heterogeneous-region or sub-cloud correspondence in Hg-I2P and SuP [2, 13, 41]; here, the two inputs already share a coordinate frame.

Separate MLP stems encode skeletal nodes and surface points; weighted first and second moments summarize each surface patch. With region-relative embeddings absorbed into f^G_{ui} and f^P_{uj}, regional aggregation is

$$g_u = \frac{\sum_i A^G_{ui} f^G_{ui}}{\max(\epsilon, \sum_i A^G_{ui})},$$
$$p_u = \frac{\sum_j A^P_{uj} f^P_{uj}}{\max(\epsilon, \sum_j A^P_{uj})}. \quad (4)$$

Region position and length codes are then added to each available modality; we reuse g_u, p_u below for these augmented features. The weights define two normalized local summaries; they do not require skeletal arclength to be numerically comparable to surface area or empirical sample mass. The aggregation is insensitive to element order when features and their incidence are permuted together, but this property alone does not establish resampling invariance of the full pipeline.

Let $m^G_u, m^P_u \in \{0, 1\}$ indicate whether the corresponding modality is present, with missing features set to zero. The fusion module produces

$$b_u = \frac{g_u + p_u}{\max(1, m^G_u + m^P_u)},$$
$$h_u = b_u + 0.1\, m^G_u m^P_u F_\theta([g_u, p_u]). \quad (5)$$

Here F_θ is a learned two-layer MLP. The coefficient is fixed, but its output magnitude is not bounded to 10% of b_u. The standard residual form is an implementation choice: the architectural change is that h_u represents two corresponding local supports before contextual propagation. The `mean_only` control removes F_θ while retaining the common-region pathway.

4.3. Context, readout, and supervision

Regions adjacent along an observed chain or meeting at a junction form a region graph. Two message-passing steps exchange local evidence. Weighted pooling then forms coarse region tokens; spatial attention with three-axis RoPE updates these tokens and writes context back to the fine regions. The pooling mass is

$$\omega_u = \frac{M_u^G}{\max(\epsilon, \sum_v M_v^G)} + \frac{M_u^P}{\max(\epsilon, \sum_v M_v^P)}, \qquad (6)$$

where $M_u^G = \sum_i A_{ui}^G$ and $M_u^P = \sum_j A_{uj}^P$. Normalizing each modality before addition keeps the context weights dimensionless. Message passing, attention, and RoPE propagate the already fused regional evidence; LitePT and MSPT provide related precedents for three-dimensional rotary encoding and local/coarse context, respectively [8, 40].

Local readout aggregates regions near a query; whole-cell readout uses the regional weights; node-pair readout combines endpoint and corridor-region evidence. Each task trains its own instance of this architectural design with its task loss. There is no new fusion-specific loss, pretraining objective, or alternating optimization procedure. The retained implementations keep the H01/NREDD hierarchy repair, the frozen FlyWire hierarchy, and the TLP directional descriptor and pair readout as documented task adaptations.

The two changes target different decisions. Equal-point-budget sampling tests the spatial allocation of the input. Regional averaging and independent-stream controls test the placement and learning of fusion; within-cell surface-content shuffling tests correspondence. Removing coarse context tests propagation beyond a region. These controls provide empirical evidence for the coupled design, not a proof that its constituent operations are independently optimal or that the complete model is invariant to density and scale changes.

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